

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

1st September 2021

Subject SA2 – Life Insurance

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
- 3. Mark allocations are shown in brackets.*

- Q. 1)** You are an Appointed Actuary of a mid-sized private life insurance company in India. The Company is promoted by a leading public sector bank in India, which currently is the exclusive bancassurance partner of the Company. In addition to bancassurance channel the Company operates through multi-distribution channel across India and has been primarily writing non-linked products comprising of Non-Par Savings and immediate annuity with-out return of premium business.

The promoter of the Company (i.e. the above-mentioned Public-Sector Bank) in its recent Board meeting has expressed its interest to sell off a portion of its stake in the Insurance Company. The promoter has received expression of interest from one of the leading Private Equity (PE) Firm in India to buy its portion of stake in the Company.

As part of the regular due diligence exercise the PE firm is provided with the following excerpts from Analysis of Surplus (AOS) of Statutory Surplus results of the immediately preceding financial year:

AOS (amount in Rs. Cr)	Non-Par Savings	Immediate Annuity
Non-economic variance:		
Deaths / Mortality	5	1
Surrender / lapse	25	0
Expenses	5	4
Economic variance:		
Investment	20	10

Note:

1. The above refers to actual over expected during the period wherein expected is calculated using the opening valuation assumptions.
2. There is a fall in yields on Government and corporate bonds at most durations during the year compared to previous year.

- i)** Explain the results as outlined in the table above and what might have led to it. (16)

The Company has also provided the Analysis of movement of Embedded Value (EV AoM) results of the same time period as below:

EV AoM (amount in Rs. Cr)	Non-Par Savings	Immediate Annuity
Non-economic variance:		
Deaths / Mortality	-2	2
Surrender / lapse	-5	0
Economic variance:		
Investment	-5	-2

Note:

1. Above EV AoM is also carried out basis the last valuation non-economic assumptions.
2. The Company carries out EV calculation as per APS 10 methodology.
3. In case of EV AoM, investment surplus is determined against the best estimate interest rate as applicable at the date of valuation.

- ii) Explain with reasons why the above EV AoM results look different from the AOS results of the Company. (9)

The transfer of stake to PE is now completed and by virtue of this transfer the PE firm has got a representative in the Board of the Life Insurance Company. The PE firm (now the shareholder) has observed that historically the Insurance Company has been growing at a rate of 10% in terms of new business premium with new business margin in the range of 10%-12% and hence have advised the Strategy Officer to come up with plan to increase both the growth rate and new business margin.

The Strategy Officer of the Company has outlined the below actionable:

Proposal 1 to increase new business growth rate: Introduce a new Unit-linked Product and replace major portion of Non-Par Savings Product new business with ULIP, as the average ticket size of ULIP plans are higher and easier to sell.

- iii) Explain the difference in policyholder taxation between Non-Par Savings life insurance plan and Unit-linked life insurance plan as per latest Income Tax provisions. (3)

- iv) As an Appointed Actuary you are requested to assess the above proposal in terms of:

- a) capital requirement (4)

- b) profitability; and (3)

- c) risk (4)

(Reference to Products regulation and solvency regulation may be provided wherever appropriate / relevant)

Proposal 2 to increase the new business margin: Strategy Officer has pointed that in the recent past other players in the market have reported healthy margin for retail protection plans and hence has suggested to introduce a new regular premium retail protection plan with competitive premium rates.

- v) Outline the key consideration when determining the basis for pricing the above product. Your response should consider the recent outbreak of COVID-19 pandemic and its implication on various bases. (11)

[50]

- Q. 2)** You are an Appointed Actuary of a mid-sized private life insurance company in India, which has only written regular premium participating and non-participating savings business in the past, with equal split of liabilities. The business written has been typically long term in nature with liabilities duration being above 25 years under both the business. The new business being written also has a similar profile and is not expected to change soon. The company expects to write significant amount of new business over the next few years. The solvency position of the company is currently 180%.

The company follows a similar investment strategy for both its participating and non-participating fund, with the following asset mix:

- 50%-55% in Government securities (GSec)
- 15%-20% in corporate bonds
- 25%-30% in equities
- 0%-5% in cash and short-term deposits

The regulator is conducting a stress test study in the industry. As part of this exercise, companies are required to project their solvency position over the next three years, under both the base scenario and stress scenario. The base scenario should be consistent with the company's business plan and best estimate assumptions.

Following are the stress tests specified by the regulator:

- Permanent reduction in 10 Yr GSec yield by 75 bps from current yield of 6.0%, and to be assumed at start of the 3-year projection period.
- Immediate fall in equity by 30%, and no recovery over the 3-year projection period.
- Permanent increase in corporate bond spreads by 50 bps, and to be assumed at start of the 3-year projection period.

The regulator has further specified that under the stress scenario, no change should be assumed against the base scenario, except to the extent of the change in the actuarial assumptions required as per the above stress tests.

- i)** Describe the cashflow projections you will carry out for this exercise, including any change required to the actuarial assumptions. (8)

(Note: you are not required to discuss the methodology of deriving any assumption)

The company has performed the stress tests as specified above and its solvency position dropped from current 180% to 125%, 120% and 115% at the end of each of the projected years.

- ii)** Discuss the possible reasons for the significant reduction in solvency position of the company. (9)

As part of the stress test exercise instructions, if a company's projected solvency position drops below 150%, the company is also required to submit an action plan which would bring its solvency to above 150%.

- iii)** Discuss what all actions can the company take to bring its projected solvency to above 150%. (10)

After looking at the stress test results, you have recommended to the CEO that the company should reduce its equity exposure from 25%-30% to 5%-10% and correspondingly increase its fixed interest securities exposure to achieve a better and stable solvency position.

The CEO agrees to your recommendation. He has however asked you if there are any considerations which needs to be taken into account before implementing the new investment strategy.

- iv)** Discuss the points which you would include in your response to the CEO. (4)

The company has a long history of declaring stable reversionary bonus rates but the terminal bonus rates have been changed frequently.

- v)** Outline the key considerations on declaration of bonus rates as per Guidance Note 6. (11)

- vi)** Discuss what potential impact can the new investment strategy have on future bonus declarations. (8)
[50]
